How synthesis choices shape supercapacitor performance

Cross-paper overview — method and material class are linked here; see Controlled Comparisons for like-for-like evidence.

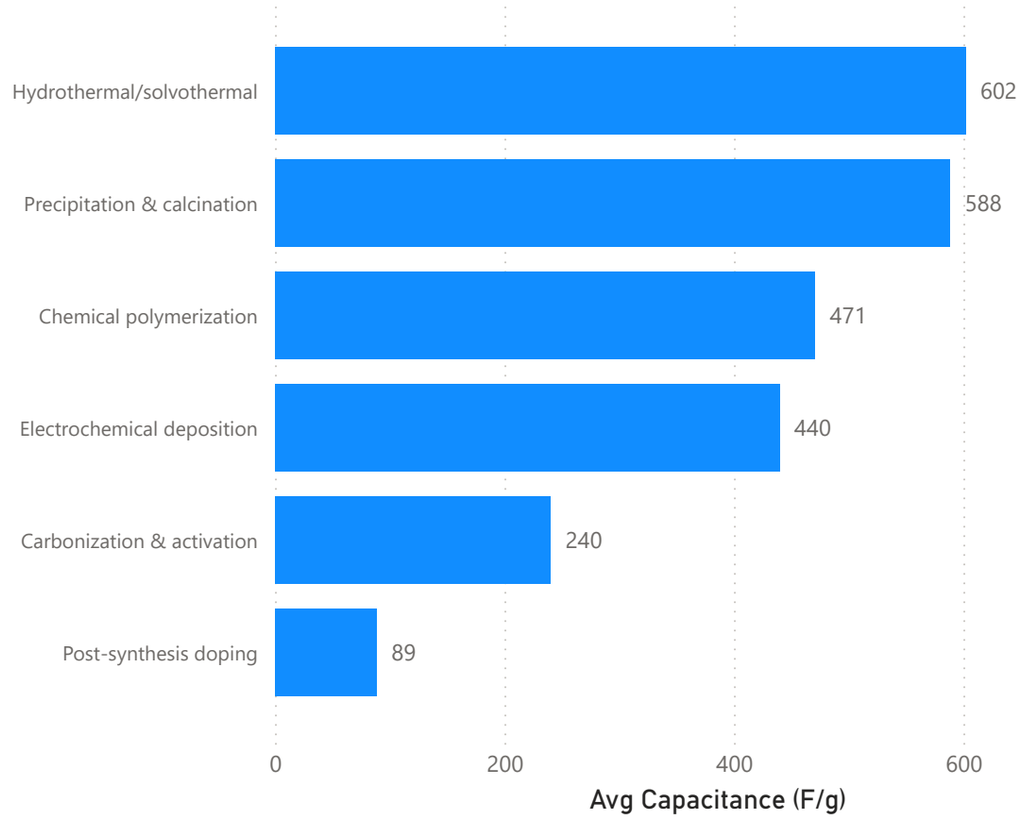
Measurement method

CV | GCD

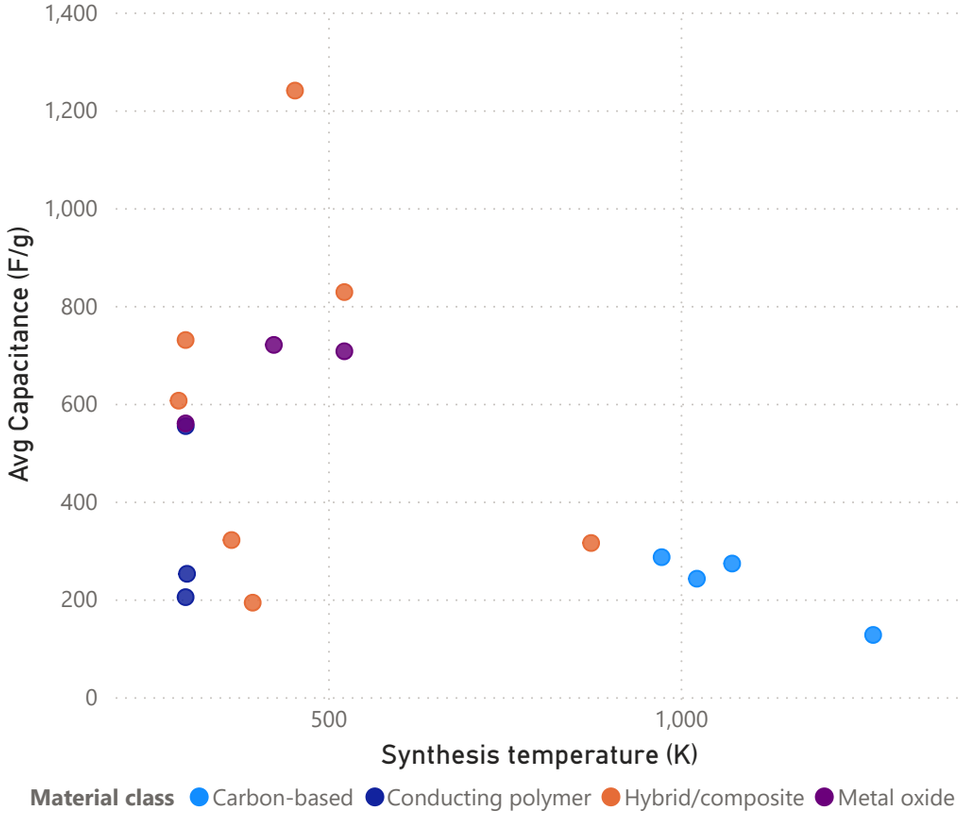
Material class

Carbon-based | Hybrid/composite | Conducting polymer | Metal oxide

Average capacitance by synthesis route



Synthesis temperature vs. capacitance

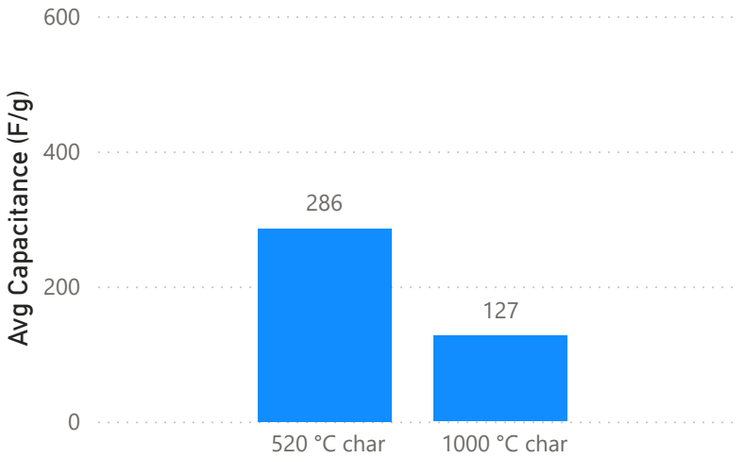


Overview | Controlled Comparisons | Explorer

Controlled comparisons: same paper, same test, one synthesis change

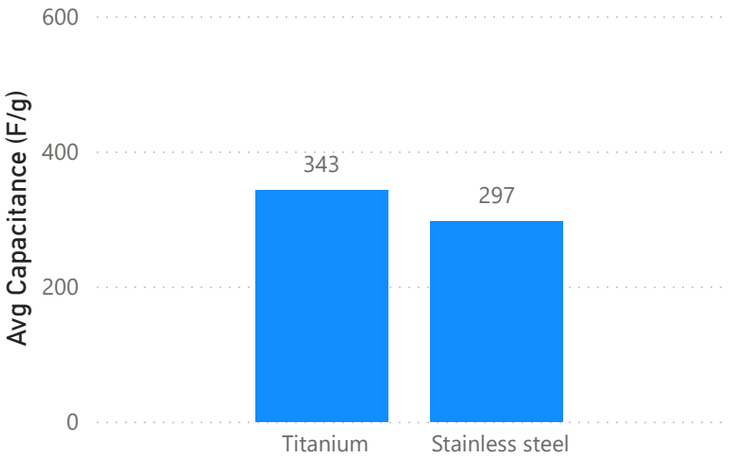
Within each chart, electrolyte and test method are identical. Values are not comparable across charts

Coal pretreatment temperature



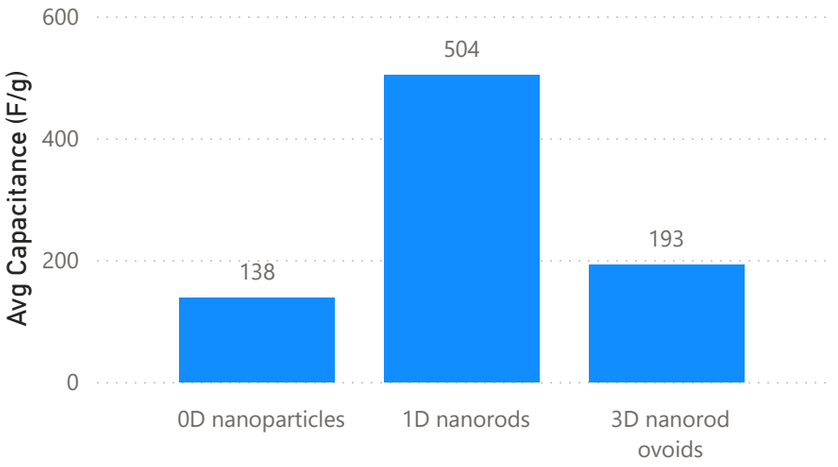
Charring at 1000 °C instead of 520 °C more than halves capacitance.

Electropolymerization substrate



Titanium gives 15% more capacitance but loses more stability (77% vs. 91% retention)

Fe_2O_3 morphology



1D nanorods reach 3.6× the capacitance of nanoparticles with identical composition

Substrate trade-off: capacitance vs. stability

Substrate	Avg Capacitance (F/g)	Cycles	Retention (%)
Stainless steel	297.00	4,000	91.00
Titanium	343.00	4,000	77.00

Overview | Controlled Comparisons | Explorer

Explorer: Every result, traceable to its source

Material class

- ☐ Carbon-based
- ☐ Conducting polymer
- ☐ Hybrid/composite
- ☐ Metal oxide

Method category

- ☐ Carbonization & activation
- ☐ Chemical polymerization
- ☐ Electrochemical deposition
- ☐ Hydrothermal/solvothermal
- ☐ Post-synthesis doping
- ☐ Precipitation & calcination

Measurement type

- ○ CV
- ○ GCD

ID	Source	Year	Material
1	Prasad, K.R.; Miura, N. Electrochem. Commun. 6 (2004) 1004-1008	2004	Nickel-Manganese Oxide (NMO)
2	Prasad, K.R.; Miura, N. Electrochem. Commun. 6 (2004) 1004-1008	2004	Cobalt-Manganese Oxide (CMO)
3	Zheng, J.P.; Cygan, P.J.; Jow, T.R. J. Electrochem. Soc. 142(8) (1995) 2699-2703	1995	Hydrous Ruthenium Oxide (RuO2.xH2O)
4	Zhao, D. et al. Polymers 15 (2023) 213	2023	Multi-Porous Carbon Nanofiber (MPCNF-750)
5	Shang, T. et al. Nano Energy 70 (2020) 104531	2020	Walnut-shell Activated Carbon (ACWS-6)
6	Shang, T. et al. Nano Energy 70 (2020) 104531	2020	Walnut-shell Activated Carbon (ACWS-6)
7	Raymundo-Pinero, E. et al. Carbon 44 (2006) 2498-2507	2006	Activated Carbon (AC520/7, from bituminous coal)
8	Raymundo-Pinero, E. et al. Carbon 44 (2006) 2498-2507	2006	Activated Carbon (AC1000/7, from bituminous coal)
9	Patil, B.H.; Jagadale, A.D.; Lokhande, C.D. Synth. Met. 162 (2012) 1400-1405	2012	Polythiophene (PTh) thin film
10	Rajesh, M. et al. Electrochim. Acta 220 (2016) 373-383	2016	Phytic acid-doped Polypyrrole (PA/PPy) on stainless steel
11	Rajesh, M. et al. Electrochim. Acta 220 (2016) 373-383	2016	Phytic acid-doped Polypyrrole (PA/PPy) on titanium
12	Sivakkumar, S.R. et al. J. Power Sources 171 (2007) 1062-1068	2007	Polyaniline (PANI) nanofibres
13	Sivakkumar, S.R. et al. J. Power Sources 171 (2007) 1062-1068	2007	Polyaniline/Multi-walled Carbon Nanotube composite (PANI/CNT, 74:26 w/w)
14	Ryu, K.S. et al. J. Power Sources 103 (2002) 305-309	2002	Polyaniline doped with HCl (Pani-HCl)
15	Ryu, K.S. et al. J. Power Sources 103 (2002) 305-309	2002	Polyaniline doped with LiPF6 (Pani-LiPF6)
16	Mayakkannan, M. et al. J. Energy Storage 44 (2021) 103257	2021	Nickel-Cobalt Ferrite (NiCoFe2O4)
17	Sohouli, E.; Adib, K.; Maddah, B.; Najafi, M. J. Energy Storage 52 (2022) 104987	2022	Carbon nano-onions/MnO2/Fe3O4 nanocomposite (CNO/MnO2/Fe3O4)
18	Yue, L. et al. Solid State Ionics 329 (2019) 15-24	2019	CoFe2O4/Carbon Nanotubes composite (CoFe2O4/CNTs)
19	Kim, B.K.; Chabot, V.; Yu, A. Electrochim. Acta 109 (2013) 370-380	2013	NiO/alpha-Ni(OH)2 hybrid flower structure (no carbon support)
20	Kim, B.K.; Chabot, V.; Yu, A. Electrochim. Acta 109 (2013) 370-380	2013	NiO/alpha-Ni(OH)2 flower with single-walled carbon nanotube support (NiO/SWCI
21	Low, Q.X.; Ho, G.W. Nano Energy 5 (2014) 28-35	2014	Fe2O3 nanoparticles + reduced graphene oxide (NP-rGO)
22	Low, Q.X.; Ho, G.W. Nano Energy 5 (2014) 28-35	2014	Fe2O3 nanorods + reduced graphene oxide (NR-rGO)
23	Low, Q.X.; Ho, G.W. Nano Energy 5 (2014) 28-35	2014	Fe2O3 self-assembled nanorod ovoids + reduced graphene oxide (ANR-rGO)